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10 – 14 March 2026, New Delhi

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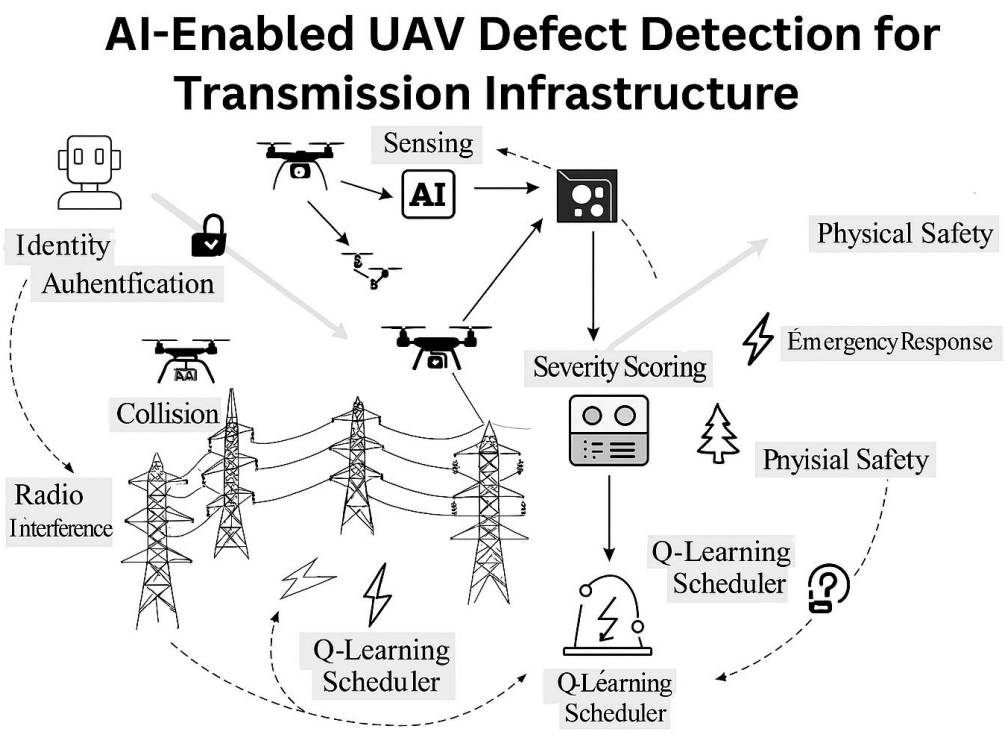
Session : Disruptive Innovations for Utilities

AI-Enabled UAV Defect Detection for Transmission Infrastructure (Ref. 26215)

Presented By

Utkarsh Jaiswal, Lead of Operations – Mechanization, Resonia Limited
Prakhar Varshney, Project Manager – Mechanization, Resonia Limited

INTRODUCTION



Limitations of Traditional Inspections

Manual inspections are slow, unsafe, and often miss subtle defects in transmission lines and that too with No real time records.

Advanced UAV Imaging Technologies

UAVs equipped with high-resolution RGB and thermal sensors capture data in large number of images from various angles to find out the accurate asset assessment.

Smart Classification of Data

Organizing bulk raw data into asset-specific segments

AI-Driven Automated Defect Detection

Analyzes classified data to detect faults, rank risk, and trigger precise, real-time interventions across transmission line corridors.

CONTEXT & BENEFITS

- **Demand for Predictive Maintenance**

  Rising power needs and digitalization drive the shift to AI-enabled UAV inspections for predictive maintenance.

- **Advances in Detection through AI**

  Enhanced small-object detection and vision-language pretraining improve inspection accuracy in complex environments.

- **Benefits of UAV-AI Systems**

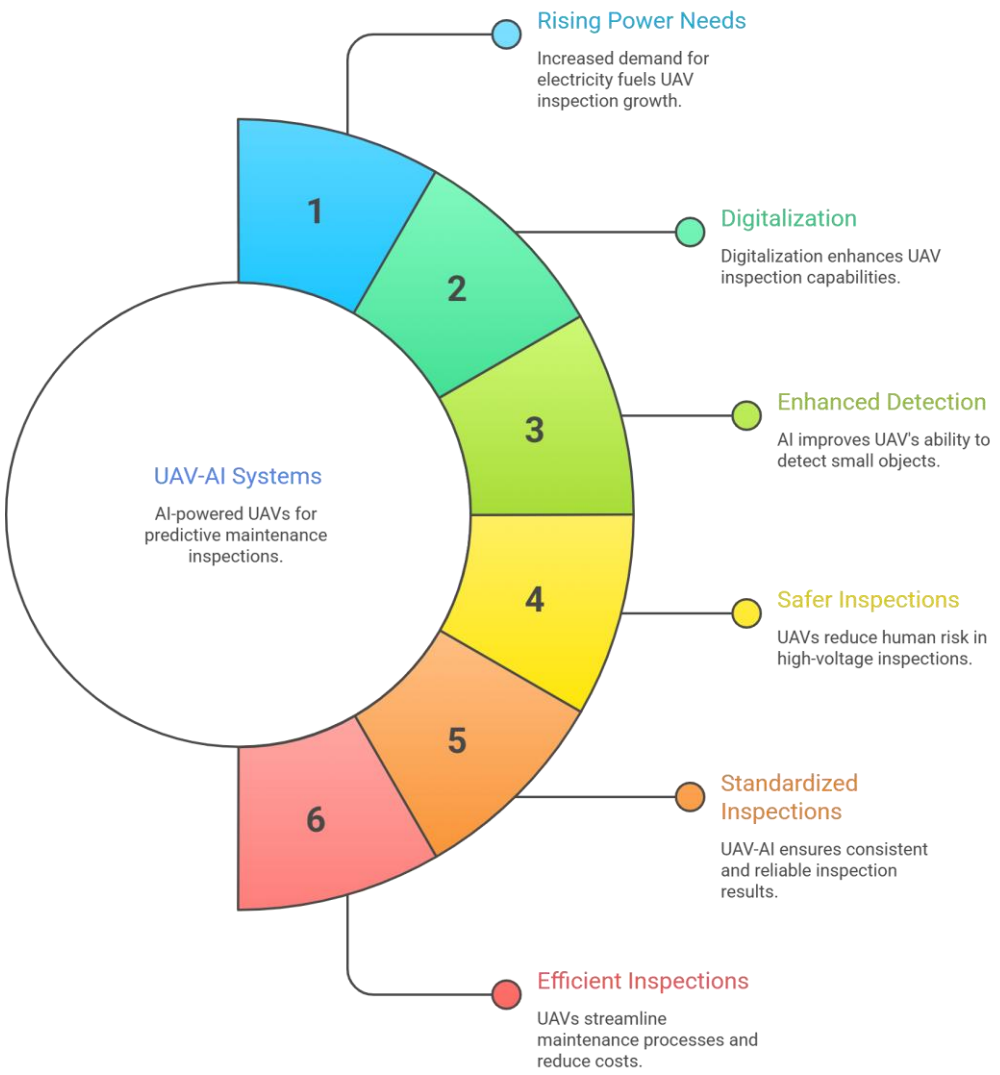
  UAV-AI systems offer safer, standardized, and efficient inspection methods for high-voltage transmission assets.

  ✓ **Automated Defect Detection**

  Automated UAV inspections minimize subjective errors, ensuring consistent evaluations in difficult terrain.

  ✓ **Data Traceability and Forecasting**

  Digital UAV data enables trend analysis, risk forecasting, and optimized asset lifecycle management.



USE / CASE STUDY - LVTPL 765 KV CORRIDOR

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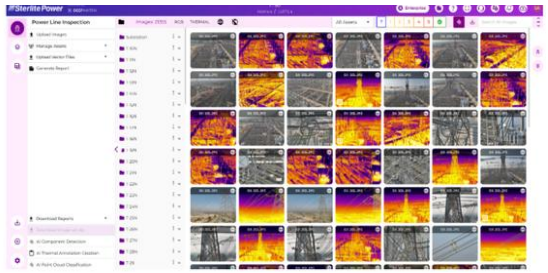
(1) - Project Synopsys

Line	Total tower
132 KV DC HEP-North Lakhimpur	92
132 KV DC Nirjuli LILO	57
LVTPL- 765kV Lakadia SS to Vadodra SS	816

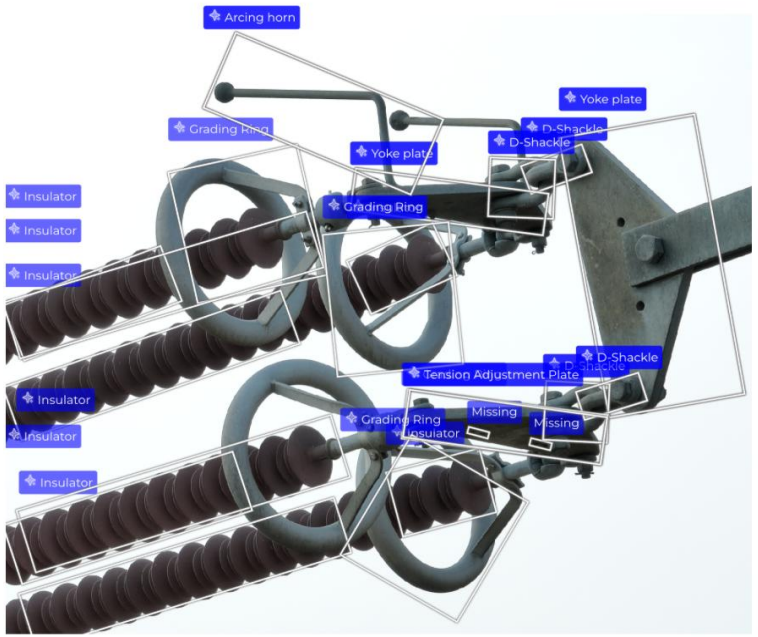
(2) - Data Collection

No of images captured overall - 4 lakh + (RGB & Thermal)

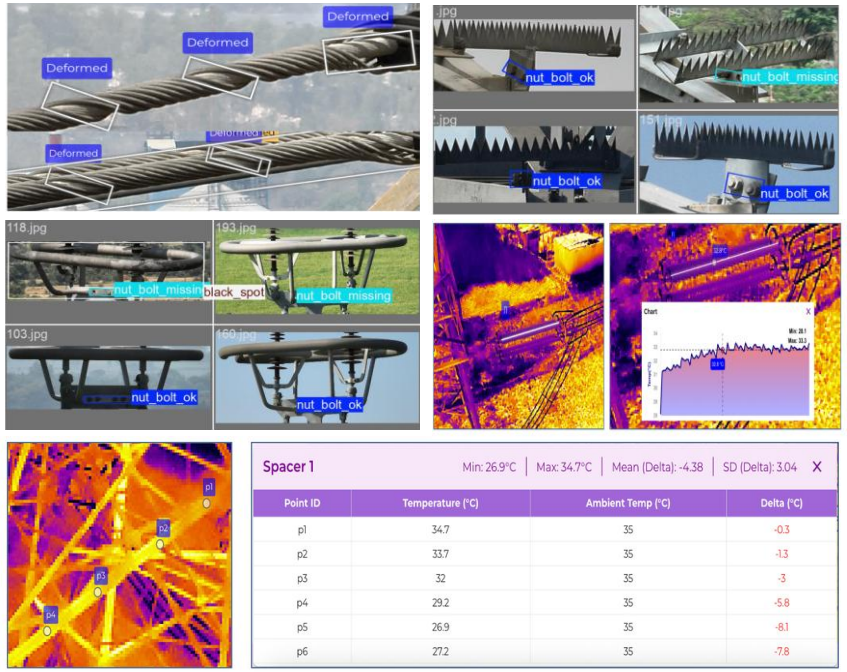
(3) - Data Classification



(4) - Comp. Identification & Annotation



(5) - Defect Detection



(6) – Report Creation

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)